

Mathematics A

Unit 2 • Chapter OL-T47 Classified

Cross-session • chapter-organised candidate

11 classified items • 42 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Unit 2 • Chapter OL-T47 • 11 items • 42 marks

Chapter	Items	Marks	Starts
01. OL-T47 Area & Volume of Similar Shapes	11	42	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Area & Volume of Similar Shapes

Unit 2 • 42 marks • 11 item(s)

June 2025 | 4MA1-2H Q26 | 4 marks

Solve a volume difference or sum for similar solids

November 2023 | 4MA1-1H Q23 | 5 marks

Convert between area and volume ratios

November 2023 | 4MA1-2H Q23 | 5 marks

Use similar solids to solve a frustum problem

June 2023 | 4MA1-1H Q23 | 5 marks

Use similar solids to solve a frustum problem

January 2022 | 4MA1-1H Q17 | 4 marks

Solve a volume difference or sum for similar solids

January 2021 | 4MA1-1H Q20 | 3 marks

Convert between area and volume ratios

June 2021 | 4MA1-2H Q20 | 3 marks

Convert between area and volume ratios

November 2021 | 4MA1-1H Q14 | 3 marks

Find an area from a known length scale factor

May-Nov 2020 | 1H Q20 | 3 marks

Solve an area difference or sum for similar figures

January 2023 | 2H Q19 | 4 marks

Solve an area difference or sum for similar figures

November 2025 | 2H Q23 | 3 marks

Solve a volume difference or sum for similar solids

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

26 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 26 is 4 marks)

WORKING SPACE

Question 18

4MA1-1H Q23 demand

MODULAR UNIT 2

5 marks

23 Solid **A** is similar to solid **B**Here is some information about solid **A** and solid **B**

	solid A	solid B
Height (cm)	3^x	
Area (cm ²)	7776	486
Volume (cm ³)	8^x	2^{x+4}

Work out the height of solid **B**
 Give your answer as a decimal.

..... cm

WORKING SPACE

Question 36

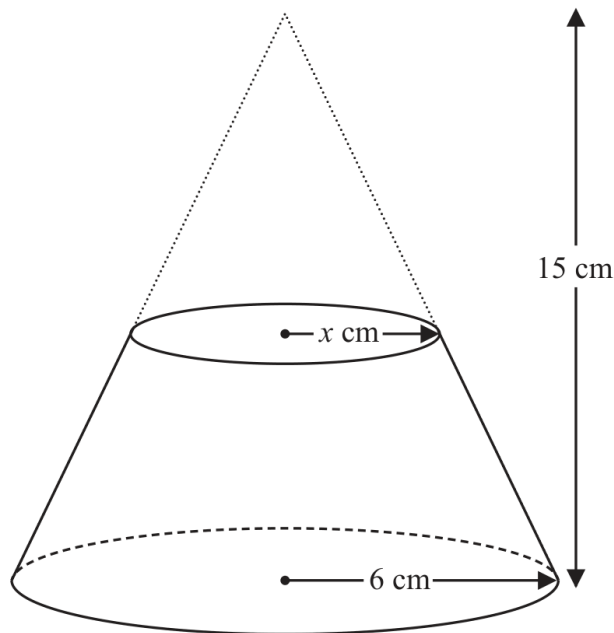
4MA1-2H Q23 whole

MODULAR UNIT 2

5 marks

23 Here is a frustum of a cone.

The frustum is made by removing a small cone from a similar large cone.



The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.

The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25}\pi \text{ cm}^3$

work out the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

WORKING SPACE

Working space for Question 36

Continue your method

UNIT 2

5 marks

WORKING SPACE

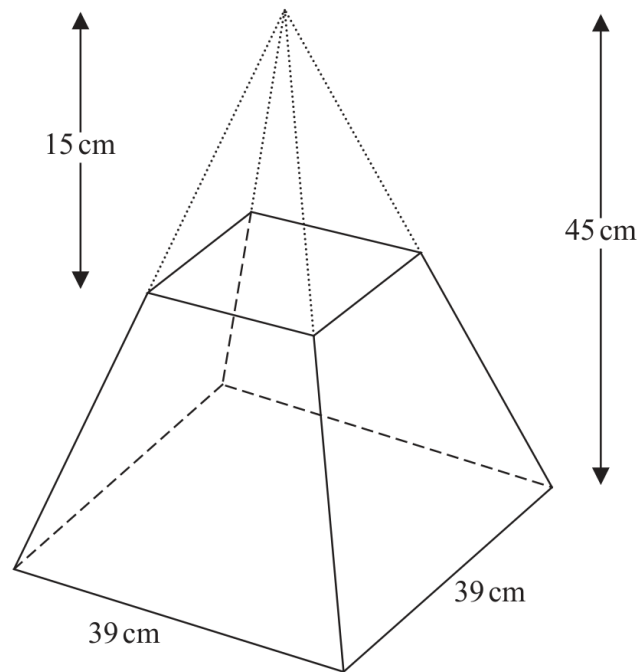
Question 18

4MA1-1H Q23

MODULAR UNIT 2

5 marks

- 23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.



The height of the small pyramid is 15 cm.

The height of the large pyramid is 45 cm.

The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.

Give your answer correct to the nearest whole number.

..... cm²

Working space for Question 18

Continue your method

UNIT 2
5 marks

WORKING SPACE

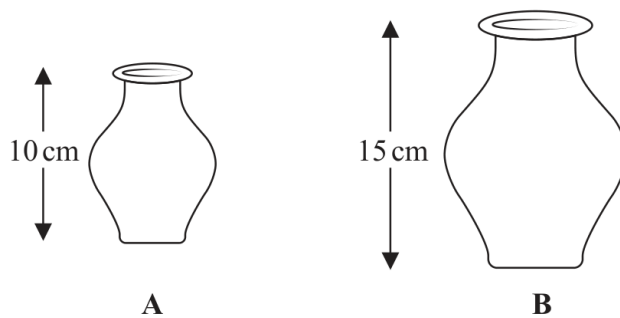
Question 11

4MA1-1H Q17

MODULAR UNIT 2

4 marks

17 **A** and **B** are two similar vases.



Vase **A** has height 10 cm.

Vase **B** has height 15 cm.

The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3

Calculate the volume of vase **A**

..... cm^3

Working space for Question 11

Continue your method

UNIT 2

4 marks

WORKING SPACE

20 **A** and **B** are two similar solids.

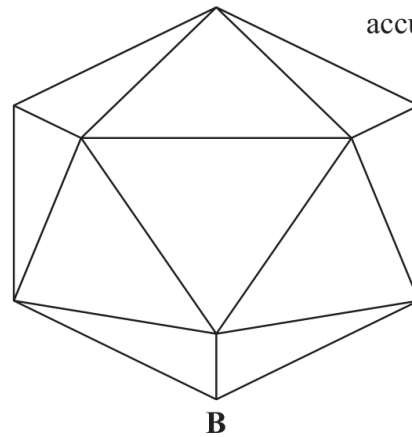
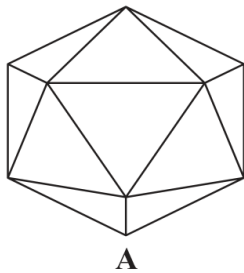


Diagram **NOT**
accurately drawn

A has a volume of 1836 cm^3

B has a volume of 4352 cm^3

B has a total surface area of 1120 cm^2

Work out the total surface area of **A**.

..... cm^2

YOUR WORKING

20 Mathematically similar wooden blocks are made in a workshop.

There are small blocks and there are large blocks.

The volume of each small block is 300 cm^3

Given that

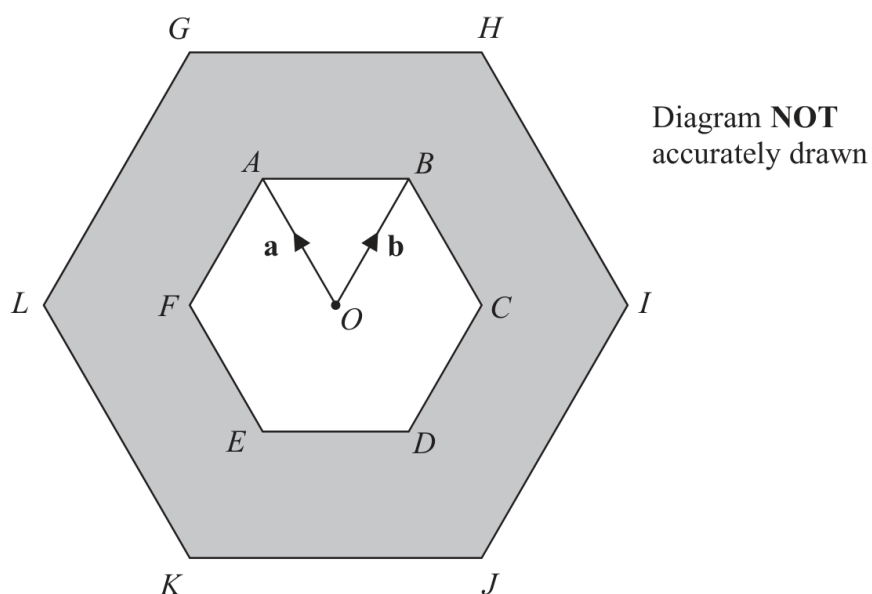
the surface area of each small block : the surface area of each large block = $25 : 36$

work out the volume of each large block.

..... cm^3

YOUR WORKING

14 $ABCDEF$ and $GHIJKL$ are regular hexagons each with centre O .



$GHIJKL$ is an enlargement of $ABCDEF$, with centre O and scale factor 2

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

(a) Write the following vectors, in terms of $\mathbf{a}$ and $\mathbf{b}$.
Simplify your answers.

(i) $\vec{AB}$

(1)

(ii) $\vec{KI}$

(2)

(iii) $\vec{LD}$

(2)

The triangle OAB has an area of 5 cm^2

(b) Calculate the area of the shaded region.

(3)

cm^2

YOUR WORKING

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area 108 cm^2 and volume 135 cm^3

Shape **S** has surface area 300 cm^2

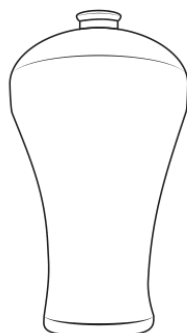
Work out the volume of shape **S**.

..... cm^3

(Total for Question 20 is 3 marks)

YOUR WORKING

19 **A** and **B** are two similar vases.



A



B

Diagram **NOT**
accurately drawn

The vases are such that

$$\text{surface area of vase B} = \frac{25}{64} \times \text{surface area of vase A}$$

and that

$$\text{volume of vase A} - \text{volume of vase B} = 541.8 \text{ cm}^3$$

Calculate the volume of vase **B**

..... cm³

(Total for Question 19 is 4 marks)

YOUR WORKING

23 Shape P is similar to shape Q

The table shows some information about shape P and shape Q

	Surface area (cm ²)	Volume (cm ³)
Shape P	200	672
Shape Q	450	

Work out the volume of shape Q

..... cm³

(Total for Question 23 is 3 marks)

YOUR WORKING